Qwen

From Diffusion Refinement to Foundation Models: A Survey of Open-Source AMT Systems (2025–2026

Generative Refinement Models for Piano Transcription

The period between 2025 and 2026 has seen a significant evolution in the application of generative models for automatic music transcription, particularly for the challenging task of solo piano. Researchers have moved beyond traditional sequence-to-sequence prediction and classification methods towards more sophisticated diffusion-based techniques that model the transcription process as a generative or refinement problem. This shift reflects a broader trend in machine learning towards leveraging generative architectures for their ability to produce high-quality, coherent outputs by capturing complex data distributions. Two prominent examples from this era are the D3RM model for piano transcription and the Noise-to-Notes framework for drum transcription, both of which utilize diffusion modeling to achieve state-of-the-art results. These approaches signify a conceptual advancement, treating transcription not merely as an inference task but as a process of generating musical notation from noise or refining a coarse approximation into a precise one.

A key development in this area is the introduction of discrete denoising diffusion models specifically tailored for the structured nature of musical data. The Discrete Denoising Diffusion Refinement Model (D3RM), introduced in a paper submitted on January 9, 2025, exemplifies this trend [3]. Unlike its predecessor, D3PIA, which was designed as a generative model requiring greater capacity, D3RM functions as a refinement model [3][9]. Its core architecture involves a denoising module that uses Neighborhood Attention layers to progressively refine a target high-resolution piano roll [3]. The model conditions this refinement process on features extracted from a pretrained acoustic model, allowing it to leverage learned representations of raw audio to correct errors in an initial transcription guess [3]. A critical innovation in the D3RM paper is the novel strategy of applying distinct transition states during the training and inference phases of the discrete diffusion model, which enhances the quality of the final refined output [3]. Experiments conducted on the standard MAESTRO dataset demonstrated that D3RM outperforms previous diffusion-based piano transcription models and establishes a new benchmark for

refinement-based approaches in terms of F1 score [3]. While the abstract confirms that the code for D3RM is available, the specific link is not detailed in the provided context [3].

Another significant contribution to generative AMT is the Noise-to-Notes (N2N) framework, presented in a paper posted to arXiv on September 26, 2025 [5]. This model redefines Automatic Drum Transcription (ADT) as a conditional generative task, where audio-conditioned Gaussian noise is transformed into a complete set of drum events, each with associated velocities [5]. This represents a direct application of diffusion modeling to a different but related sub-task within music transcription. A primary challenge in this domain is the joint optimization of binary onset detection and continuous velocity estimation, which can be difficult for standard diffusion models to handle effectively. To address this, the N2N method employs an Annealed Pseudo-Huber loss function, which facilitates the simultaneous and effective training of these two distinct output types [5]. Furthermore, to improve robustness on out-of-domain drum audio—a common issue in real-world applications—the framework incorporates high-level semantic features extracted from Music Foundation Models (MFMs). These features augment the low-level spectrogram inputs, providing the model with a richer contextual understanding that helps disentangle overlapping percussive sounds [5]. The experimental results demonstrate that the N2N model achieves a new state-of-the-art performance across multiple ADT benchmarks, validating the efficacy of this diffusion-based generative approach [5]. The author of this work is Michael Yeung [5].

These generative models, D3RM and N2N, share a common philosophical underpinning: they treat transcription as a generation or refinement process rather than a simple classification. By using diffusion models, they implicitly learn the joint probability distribution of all possible note events or drum hits, which can lead to more musically coherent and structurally consistent transcriptions, especially in passages with dense polyphony or complex rhythmic patterns where autoregressive models might falter due to error propagation. The success of these models suggests that the future of high-fidelity transcription may lie in these generative paradigms, which offer a powerful alternative to purely discriminative or autoregressive architectures. The focus on discrete diffusion for piano roll refinement and conditional generation for drum events highlights the adaptability of these models to the specific structures inherent in different musical domains.

Foundation Model Architectures for Multi-Instrumental Transcription

In parallel with advancements in generative modeling, a dominant theme in the 2025-2026 landscape of automatic music transcription is the widespread adoption and adaptation of large-scale foundation models, particularly those built upon the Multi-train, multi-domain, multi-style (MT3) paradigm. The MT3 architecture, developed by Google, has become a de facto standard and a foundational baseline for many state-of-the-art systems, especially for the notoriously difficult problem of multi-instrumental transcription (MIT). Its core strength lies in its ability to process diverse musical inputs and generate corresponding symbolic scores, making it an ideal starting point for tackling polyphony and cross-instrument generalization. Analysis of recent top-performing entries in the 2025 AMT Challenge reveals that nearly all successful models were built by extending or augmenting the existing MT3 framework, indicating that architectural innovation is now focused on enhancing this proven blueprint rather than replacing it entirely [7].

The winning system of the 2025 AMT Challenge, named MIROS, provides a prime example of this trend [7]. MIROS is based on the YourMT3+ encoder-decoder framework, a variant of MT3, but introduces several key enhancements. Most notably, it replaces the standard encoder backbone with a MusicFM conformer-based self-supervised foundation model [7]. This substitution leverages a more powerful and specialized representation of the input audio, potentially leading to better feature extraction and improved transcription accuracy. The rest of the architecture follows the YourMT3+ design, utilizing T5-style decoders equipped with cross-attention mechanisms, rotary position embeddings, and FlashAttention for efficient hardware utilization [7]. A crucial component for handling polyphony is the recurrent adapter, which conditions the decoder's outputs on learned instrument group embeddings. This allows the model to dynamically adjust its predictions based on the expected instrumentation within a given segment of music, effectively managing the complexities of multi-part textures [7]. The success of MIROS, which achieved an F1 score of 0.5998 in the challenge, underscores the power of combining a robust foundational architecture like MT3 with advanced encoders and specialized modules for instrument separation [7].

The second-place model in the same challenge, 'YourMT3-YPTF-MoE-M', further illustrates the strategy of augmenting the MT3 paradigm [7]. This model extends the MT3 architecture with two significant additions: Mixture-of-Experts (MoE) routing and hierarchical time-frequency attention [7]. MoE is a technique that allows a model to have

a large number of parameters while keeping computational costs manageable during inference by activating only a subset of "expert" networks for any given input. This modular approach can enhance the model's capacity to handle diverse musical styles and instruments. Hierarchical time-frequency attention, meanwhile, enables the model to capture dependencies at multiple scales, from fine-grained rhythmic details to broader harmonic structures, which is essential for accurate transcription of complex pieces. The third-place model, 'YourMT3-YPTF-S', also builds upon the MT3 architecture but incorporates different innovations, such as self-supervised random projection quantization and auxiliary onset/offset losses, to further boost performance [7]. Together, these top three models demonstrate a clear consensus: the MT3 framework provides a highly effective and flexible foundation upon which researchers can layer increasingly sophisticated components to push the boundaries of what is possible in multi-instrumental transcription.

This reliance on the MT3 paradigm has important implications for the field. It suggests that the fundamental challenges of MIT—namely, separating overlapping sound sources and accurately predicting note timings across multiple instruments—are being addressed effectively by this general-purpose architecture. Consequently, the frontier of innovation is shifting towards three main areas: the choice of the encoder (e.g., using specialized foundation models like MusicFM), the integration of specialized modules for tasks like instrument separation (e.g., recurrent adapters), and optimizations of the training process itself (e.g., novel loss functions, MoE routing). However, this heavy reliance also points to a potential bottleneck. The fact that nearly all top models use a common training corpus of ten datasets indicates that progress may soon be constrained less by architectural differences and more by the availability of diverse, high-quality labeled data [7]. The challenge of data scarcity remains a critical factor limiting the generalization capabilities of even the most advanced foundation model-based systems.

Lightweight and Specialized Architectures for Polyphony

While foundation models offer immense power, their complexity and resource requirements can be prohibitive for real-world applications, especially on consumer-grade hardware. In response, a parallel line of research in 2025-2026 has focused on developing lightweight yet effective architectures specifically designed to tackle the core challenge of polyphony in multi-instrumental transcription. These models prioritize efficiency and practical deployment without sacrificing competitive performance. One of

the most notable examples in this category is `timbreAMT`, a model that introduces a novel "note-level" deep clustering approach to perform dynamic instrument separation. This innovative strategy, combined with a lightweight two-branch architecture, makes it suitable for low-resource environments while achieving strong results on real-world polyphonic datasets [4].

The `timbreAMT` model proposes a two-branch deep learning architecture that extends a timbre-agnostic transcription backbone, inspired by BasicPitch [4]. The first branch predicts frame-level activations, similar to a standard transcription model. The second, dedicated branch learns direction-aware embeddings for each instrument class, providing the model with explicit information about the timbral characteristics of the sources it needs to separate [4]. The key innovation lies in the application of deep clustering at the 'note level' rather than the more traditional 'frame level'. Instead of clustering individual time-frequency bins in a spectrogram—which can be computationally expensive and prone to fragmentation—the model first aggregates adjacent bins into coherent note events. These aggregated note events are then clustered, which significantly reduces the computational overhead and produces cleaner, more structured instrument separations [4]. This approach mitigates the fragmentation often seen in frame-level clustering, where a single note can be incorrectly split across multiple clusters. To further enhance performance and stability, the model employs technical optimizations such as EnergyNorm for spectral normalization, dilated convolutions to capture a wider harmonic context, and an InfoNCE loss function for more efficient cluster formation compared to MSE-based alternatives [4].

The practical benefits of this lightweight design are substantial. On an Intel® Ultra 7 255H CPU, the `timbreAMT` model transcribes a 301-second duet in just 17.1 seconds, which is significantly faster than the 39.1 seconds required by the BasicPitch baseline [4]. Despite its efficiency, the model demonstrates competitive performance on challenging real-world polyphonic datasets, including BACH10, URMP, and PHENICX. This balance of speed and accuracy makes it a valuable tool for applications where real-time or near-real-time processing is desired. The model was trained on a combination of datasets, including MusicNet, BACH10, URMP, PHENICX, and a custom synthetic dataset, showcasing a diverse training regimen that likely contributes to its robustness [4]. The entire project, including source code, training scripts, and model weights, is publicly available on GitHub, ensuring reproducibility and accessibility for the research community [4].

Other approaches from the same period also address polyphony through specialized architectural choices. The winning MIROS model from the 2025 AMT Challenge, for

instance, uses a recurrent adapter to condition its outputs on learned instrument group embeddings [7]. This allows the model to implicitly manage polyphony by dynamically adapting its predictions based on the expected instrumental configuration. Similarly, the Noise-to-Notes (N2N) framework for drum transcription improves robustness on out-of-domain audio by incorporating features from Music Foundation Models (MFMs) [5]. These higher-level semantic cues help the model better understand the musical context, which aids in disentangling overlapping drum hits. The collective effort of these diverse solutions—note-level clustering in `timbreAMT`, adaptive conditioning in MIROS, and semantic augmentation in N2N—highlights that there is no single solution to the polyphony problem. Instead, the research community is exploring a multifaceted strategy that combines explicit separation techniques with implicit, context-aware generation methods. This ongoing exploration is critical, as studies have shown that model performance degrades significantly with increased polyphony; for the winning MIROS model, the F-measure dropped by over 0.28 points when scaling from one to three instruments, underscoring the persistent difficulty of this challenge [2] [7].

Open-Source Implementations and Researcher Information

A defining characteristic of the automatic music transcription research landscape from 2025 to the present is a strong commitment to open science and reproducibility, with many cutting-edge projects being released with public code repositories. This practice is invaluable for the research community, as it allows for the verification of results, fosters collaboration, and enables developers to integrate state-of-the-art models into practical applications. The provided sources highlight several key open-source projects from this period, along with the associated researchers who have made these contributions. This section compiles the available information on these projects, including their GitHub links and the contact details of their creators where identifiable.

The table below summarizes the key open-source projects discussed in the provided materials, detailing the ML/DL approach, the GitHub repository URL, and the available researcher information. This compilation serves as a direct reference for accessing and engaging with the latest advancements in the field.

Project Name	ML/DL Approach Summary	GitHub Repository	Researcher(s) & Contact
D3RM	Discrete Denoising Diffusion Refinement Model for Piano Transcription. Uses Neighborhood Attention layers to refine a piano roll conditioned on features from a pretrained acoustic model. [3]	Code is available but the specific link is not provided in the source material.	Hounsu Kim (Author) [3]
timbreAMT	Lightweight two-branch architecture using "note-level" deep clustering for dynamic instrument separation. Extends a timbre-agnostic backbone with a dedicated timbre encoder. [4]	https://github.com/madderscientist/timbreAMT [4]	MaddScientist (GitHub username) [4]
MIROS	Winner of the 2025 AMT Challenge. Based on the YourMT3+ framework with a MusicFM conformer-based encoder and a recurrent adapter for instrument group embeddings. [7]	Not Available	Authors not specified in detail.
Noise-to-Notes (N2N)	Conditional generative model for Automatic Drum Transcription using diffusion modeling. Incorporates an Annealed Pseudo-Huber loss and features from Music Foundation Models (MFMs). [5]	Not Available	Michael Yeung (Author) [5]
Transcoda	End-to-end Optical Music Recognition (OMR) system using a vision-encoder-decoder (ConvNeXt-V2 + Transformer) to transcribe sheet music images into Humdrum `kern` format. [6]	https://github.com/btrkeks/transcoda [6]	btrkeks (GitHub username) [6]
mt3-infer	A Python package for easy inference with various checkpoints of the MT3 model, defaulting to the YPTF.MoE+Multi (noPS) checkpoint. [11]	Available on PyPI [11]	Not Available

Several projects stand out for their comprehensive open-source releases. The `timbreAMT` model, developed by the user "madderscientist," is fully available on GitHub, complete with source code, training scripts, and a DOI for enhanced reproducibility [4]. Similarly, the `Transcoda` OMR system, created by "btrkeks," is released on GitHub and includes its code, training scripts, and model weights, providing a complete toolkit for optical music recognition [6]. The existence of these detailed repositories is a significant boon to the community. Additionally, the `mt3-infer` package, available on the Python Package Index (PyPI), lowers the barrier to entry for users who wish to experiment with the powerful MT3 models without needing to manage complex installations or training pipelines [11].

For other notable works, the information is less complete. The `D3RM` paper confirms that the code is available, but the specific URL is omitted from the provided abstract snippet [3]. Likewise, while the `N2N` paper identifies Michael Yeung as the author, the corresponding GitHub repository link is not provided [5]. The winning `MIROS` model from the 2025 AMT Challenge is well-documented in terms of its architecture, but the authors and their affiliations are not detailed in the provided text [7]. Despite these gaps, the overall trend is positive, with a clear emphasis on making the tools and technologies developed through this research publicly accessible. The collection of GitHub usernames

provides a direct pathway for interested parties to follow the ongoing work of these researchers and contribute to the open-source ecosystem.

Key Challenges and Future Directions in AMT

Despite the significant progress in automatic music transcription observed between 2025 and 2026, the analysis of recent publications and competition results reveals persistent challenges that define the current frontier of research. The most formidable obstacle remains the accurate transcription of dense polyphony, where multiple instruments play simultaneously. Even the most advanced models, including those based on powerful foundation architectures like MT3, exhibit substantial performance degradation as the number of concurrent instruments increases [7]. This limitation is not merely a matter of incremental improvement but points to a deeper issue in how current models perceive and disentangle overlapping sound sources. Furthermore, a systematic analysis of model performance revealed that dynamics estimation is more vulnerable to musical variation than onset prediction, and that systems suffer significant F1 score drops due to genre changes and other musical factors, suggesting a lack of true generalization [2]. These findings collectively indicate that while the field has made remarkable strides, the path forward requires addressing fundamental issues of model robustness, data diversity, and the inherent complexity of musical perception.

One of the most striking insights from recent evaluations is the profound impact of polyphony on model accuracy. A statistical analysis of the 2025 AMT Challenge results showed that the winning MIROS model's F-measure dropped by over 0.28 points when the number of instruments in a piece increased from one to three [7]. This dramatic decline highlights that dense polyphony is a primary bottleneck for current AMT systems. While innovative architectural solutions like the note-level clustering in `timbreAMT` and the recurrent adapters in MIROS represent important steps forward, they do not completely solve the problem [4] [7]. The continued reliance on these specialized techniques suggests that a more fundamental breakthrough in source separation or event detection within complex auditory scenes is needed. The challenge is compounded by the fact that nearly all top-performing models in the 2025 AMT Challenge were trained on the same limited corpus of ten datasets, indicating that data scarcity is a major constraint on progress [7]. Without access to a much larger and more diverse collection of professionally recorded, multi-instrumental music, models will continue to struggle with musical variations they have not explicitly encountered during training.

This leads directly to the second major challenge: model generalization and corpus bias. A systematic analysis conducted by Lukáš Samuel Marták and submitted to arXiv in December 2025 investigated the performance of AMT models under distribution shifts related to genre, dynamics, and polyphony [2]. The study found that a note-level F1 score drop of 20 percentage points could be attributed to sound-related factors, with an additional 14 points lost due to genre changes alone [2]. This demonstrates that current models are brittle and highly sensitive to musical variations that are trivial for human listeners. The experiments with randomly generated, non-musical sequences provided clear evidence of the "Corpus Bias" problem, where models excel within the distribution of their training data but fail when faced with novel or extreme variations [2]. This finding implies that future progress cannot rely solely on building larger and more complex models; instead, there is a pressing need to develop models that are inherently more robust to distribution shifts or to create vastly more diverse and representative training corpora.

Looking ahead, the research trajectory appears to be moving in several promising directions. First, the successful application of diffusion models in projects like D3RM and N2N suggests that generative approaches will continue to be a fertile area of exploration [3] [5]. These models may hold the key to producing more structurally coherent transcriptions by learning the joint distribution of musical events. Second, the trend of augmenting powerful foundation models like MT3 with specialized components—such as advanced encoders (MusicFM), Mixture-of-Experts, and recurrent adapters—will likely persist [7]. This hybrid approach allows researchers to build upon a proven and versatile base while adding targeted capabilities to address specific weaknesses, such as polyphony. Third, the development of lightweight, efficient models like `timbreAMT` indicates a growing awareness of the need for practical, deployable systems that can run on consumer hardware [4]. Finally, the strong commitment to open-source releases ensures that the community can collaboratively tackle these grand challenges, building upon shared tools and datasets to push the boundaries of what is possible in automatic music transcription. The ultimate goal remains the creation of a system that can match the perceptual abilities of a human musician, accurately transcribing any piece of music regardless of its complexity, style, or recording quality.

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